ESPI



TALENT IN THE EUROPEAN SPACE SECTOR

BRIDGING GAPS, BUILDING FUTURES

EXECUTIVE SUMMARY

ESPI European Space Policy Institute

TALENT IN THE EUROPEAN SPACE SECTOR

Europe is facing increasingly critical talent deficits. In 2024, the European Commission (EC) acknowledged that skills shortages and recruitment problems have been on the rise in member states for the last decade, with a deficit of skills **enabling technological and green developments**, and demographic shifts given as main reasons.[1] Indeed, just last year, a Eurobarometer survey prepared for the European Year of Skills revealed that nearly 4 out of 5 small and medium-sized enterprises (SMEs) find it hard to find workers with the right skills, and more than half having difficulties in retaining such skilled workers.[2]

> *"The European Union is facing a significant challenge in meeting the demand for skilled talent in STEM fields. These are crucial for growing strategic sectors like clean and circular technologies, digital technologies, aerospace, and defence."*
> **European Commission - DG EAC (10.6.25)**

The space sector, a key strategic enabler of innovation, economic growth, and autonomy, is no exception to such difficulties, with a 2023 OECD report emphasising high competition from other high-technology sectors as a determining factor. It highlighted that the supply of graduates in space-related and STEM disciplines does not meet demand in several OECD countries.[3] As Europe is expected to ramp up investment in space programmes to meet pressing policy objectives, the **hiring demand for skilled professionals is on an upward curve**, further exacerbating the existing talent shortfall.

New and revamped efforts to develop and foster talent do exist, i.e. the European Commission's Space Career Launchpad, ESA's acclaimed Graduate Trainee Programme, and industry-led initiatives (e.g. TAS Young Talent Programme, EnduroSat's Space Challenges Bootcamp). Yet, an understanding of talent supply and demand dynamics, trendlines, and gaps across the full value chain of the space sector is limited, relying on ad-hoc studies, and partial coverage.

ESPI's newest report, Talent in the European Space Sector: Bridging Gaps, Building Futures, provides an analysis of just under 3000 vacancies across a 6-month period, to determine which skillsets, experiences, and profiles are most needed by a diverse sample of industry actors across Europe. This was complemented by answers of over 40 HR professionals. The report also analyses over 500 answers by students and young professionals breaking into the sector, on their expectations, internal motivation and backgrounds.

The study represents a first step towards the ambition of developing a comprehensive talent observatory, aimed at providing insights and informed recommendations on talent development to Europe's institutions and decision-makers. ESPI is open to cooperating with external partners in these efforts.

Developing futureproof workforce policies, academic curricula, and upskilling programmes through informed and long-term analysis is critical at a time when Europe needs 21st century skills to bridge the innovation and competitiveness gap, ensure the continent's peace, and keep on track with decarbonisation targets.

With shifts in the US, including cuts to science research funding, the landscape is favourable to further establishing Europe as a global talent hub across the Deep Tech landscape, including space. The appropriate links and understanding between academia, business, and public policy communities will however be essential to understand and address Europe's current barriers of progress.

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[1] European Commission, "Commission sets out actions to tackle labour and skills shortages", 20 March 2024 (Link)
[2] Ipsos European Public Affairs for the EC Directorate-General for Employment, Social Affairs and Inclusion, Flash Eurobarometer 529 - European Year of Skills: Skills shortages, recruitment and retention strategies in small and medium-sized enterprises. EC, 2023 (Link)
[3] Organisation for Economic Co-operation and Development. OECD Competition Trends 2023. OECD Publishing, 2023 (Link)

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CONSIDERATIONS FOR THE BROADER TALENT ECOSYSTEM IN EUROPE

This report is designed as an analytical data resource for all stakeholders involved or concerned about the future of talent in the European space sector. However, ESPI identified some key recommendations during the analysis, that are presented below:

1. Consistent tracking of job collection dynamics across Europe, and integration into educational and industrial policy development at supranational and national level, facilitating new and improved space education initiatives that align and adapt with the sector's needs.

2. Further quantitative impact assessments of the power of inspiration in attracting the general public to the space sector, in line with the Principia mission, for both exploration and science endeavours.

3. Strengthen partnerships to develop talent pipelines between universities, industry, and space agencies, as a key element of a robust industrial policy. These stakeholders should jointly identify current and future skill gaps and co-develop the appropriate curricula, while also getting directly involved in knowledge transfer.

   - Students wishing to upskill in a particular area should also be able to enrol in modules from a wide range of education providers.

4. When defining new programmes and flagship initiatives, ensure ring-fenced budget lines devoted to the hiring and development of new talent, to ensure long-term programmatic continuity and a strong workforce base to last.

Bridging gaps and building futures for the next generation of space talent is not an isolated task for one stakeholder community but requires coordinated efforts from all stakeholders across the ecosystem, which could include the below:

The establishment of a European-wide network of HR professionals across the full value chain of space sector companies would enable the sharing of information, best practices and solutions to common recruitment challenges.

Establishing the barriers of progress, i.e. a detailed understanding of the blocking points, and associated costs to unblock them, that prevent the formation of new practices and programmes, and cross-domain coordination. European stakeholders need a forum to discuss and find solutions to such barriers together.



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